

CHRISTIAN MACOMBER

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EDUCATION

DigiPen Institute of Technology — Redmond, WA
Bachelor of Arts in Computer Science in Game Design
Fall 2024 - Spring 2028 (Expected)

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Verse, ARM Assembly
Engines & Development: Unreal Engine 5, Unity 6, Custom Game Engines, Gameplay Programming
Gameplay Systems: AI / Enemy Behavior, Game Mechanics, Gameplay Systems
Data Structures & Algorithms: Hash Tables, Graphs, AVL Trees, Binary Trees, Big-O Analysis
Production Tools: Perforce, Git/GitHub, SVN, JIRA, Visual Studio
AI Accelerated Development Tools: GPT-5.6 Sol, Claude Sonnet 5, Claude Fable 5, Kimi K3
Development Practices: Debugging, Performance Optimization, Testing, Iteration, Iterative Development, Problem Solving, Team Collaboration

PROJECTS

Horror Game | Unreal Engine 5

Team of 8, September 2026 - June 2027 (Targeted Steam release date)

- Collaborate with programmers, designers, and artists to develop a team-based game using UE5.
- Developed an enemy AI system using behavior trees to create complex and flexible enemy behaviors.
- Work with the team to test and improve gameplay features as development continues.

Top-Down Adventure Game | C++ Custom Engine

Team of 4, September 2025 – June 2026 (coming soon to steam)

- Developed gameplay systems for a top-down adventure game featuring a graveyard level and boss encounter, including item, inventory, and collision systems.

- Designed and implemented enemy AI using a behavior tree with customizable attacks, allowing multiple enemy types and a boss to share the same system while still having unique behaviors.
- Ran playtests to evaluate game flow and level design, then adjusted enemy encounters based on feedback.

Casino Game Simulator Engine | C++ Custom Engine

Team of 5, January 2025 – June 2025

- Developed a modular and extendable engine to support a variety of casino games.
- Implemented core gameplay mechanics for various card games, including poker rules and game logic.
- Implemented and tested gameplay systems while working within a custom game engine.
- Debugged gameplay behavior and refined systems through iterative testing.

Custom Game Controller | ARM Assembly

- Built a functional custom game controller combining basic circuitry with low-level input programming.
- Programmed controller input behavior in ARM Assembly and tested hardware/software interaction.

EXPERIENCE

Youth Game Development Instructor | Open World — Redmond, WA

July 2026 – August 2026

- Taught students ages 8–13 robotics engineering and programming using robotics kits and MakeCode.
- Taught students ages 6–8 introductory game development using MakeCode Arcade, guiding them through the design and programming of interactive games.
- Guided students through testing, troubleshooting, and iteration while encouraging creativity, teamwork, and problem-solving.

Computer Science Teacher Assistant | Redmond, WA

September 2025 – Present

- Assist students with introductory C programming, program logic, debugging, and foundational computer science concepts.
- Help students diagnose programming errors and develop systematic problem-solving and debugging strategies.
- Explain programming and game development concepts through one-on-one technical support.
- Provide clear and timely feedback on programming assignments and coursework.

Additional Experience (2021-Present): Amazon, Subway, and DigiPen — additional roles demonstrating teamwork, reliability, adaptability, and experience working in collaborative environments.